

HAFSA SHAHID

AI Engineer

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AI Fullstack Engineer shipping LLM-powered applications, autonomous agents, and n8n automation pipelines to production. Proficient in TypeScript, Python, Next.js, FastAPI, and PostgreSQL, with deep hands-on experience in RAG, prompt engineering, agentic workflows, and multi-platform API integration.

TECHNICAL SKILLS

Languages: TypeScript, Python, Java, C

AI & LLMs: Claude API, GPT-4o, Prompt Engineering, RAG, Embeddings, Evals

Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3

Backend & DB: FastAPI, Node.js, PostgreSQL, Supabase (pgvector), Qdrant

Automation: n8n, Playwright, Zoho Mail API, Google Workspace APIs, OAuth 2.0, agentic coding workflows

DevOps: Docker, Git, CI/CD, PM2, Runpod, Vercel, Railway, Linux

WORK EXPERIENCE

Junior AI Engineer · VK Agents

Oct 2025 – May 2026

- Shipped features for a production Next.js app in agile sprints; maintained unit test suite, enforced code coverage, and managed Git contributions via PR-based branching workflow.
- Practiced spec-driven development using Claude Code as the primary agentic coding tool; maintained CONTEXT.md and DECISIONS.md for architectural continuity across sessions.
- Led comprehensive website revamps and optimizations, enhancing user experience, site speed, and technical functionality across multiple platforms.
- Developed and fine-tuned voice agent models and pipelines, migrating projects to Runpod for GPU-accelerated performance and reduced latency.
- Established and optimized Fiverr and LinkedIn strategies, including gig creation, carousel automation, and promotional campaigns, boosting brand visibility and lead generation.
- Scoped and delivered AI automation solutions for B2B clients end-to-end, from requirements and system design through n8n orchestration, API integration, and deployment.
- Managed client-facing deliverables including proposal pipelines, outreach systems, and voice agents, coordinating between technical implementation and business requirements.
- Debugged and iterated on production webhook integrations, OAuth token flows, and multi-step automation pipelines; resolved edge cases in thread continuity, duplicate detection, and API rate handling.

PROJECTS

Full-Time Employee: Autonomous AI Agent

Python · Claude Code · Gmail API · LinkedIn/Twitter/Meta APIs · Playwright · Odoo CRM · PM2 · Docker · Obsidian

- Built a perception-planning-execution autonomous agent: PM2-managed Python watchers monitor Gmail and file drops, feed a Claude Code planning engine that classifies inputs and drafts action plans, and sync state to an Obsidian vault.
- Gated all outbound actions behind a file-based human-in-the-loop inbox; dispatched execution across six platforms (Gmail, LinkedIn, Twitter/X, Facebook, Instagram, Odoo 19 CRM) via Playwright and native API clients, with JSONL audit logging and draft-only invoice creation to prevent unintended actions.

AI-Powered B2B Proposal Engine

n8n · GPT-4o · GPT-4o-mini · OpenAI Embeddings · Supabase (pgvector) · Google Slides API · Google Drive API · Gmail API

- Built a RAG-powered proposal generation pipeline in n8n:** webhook ingests a client brief → GPT-4o-mini classifies metadata → brief is embedded and matched against a Supabase pgvector store of past proposals → top-3 results injected as context → GPT-4o outputs a structured JSON proposal.
- Automated end-to-end document delivery:** GPT-4o output populates a copied Google Slides template via batchUpdate API, exported to PDF via Drive API, and uploaded to Drive; a Gmail approve/reject webhook gates final delivery with zero manual formatting.

Advanced B2B Cold Email Outreach Automation

n8n · Zoho Mail API · Google Sheets · OAuth 2.0

- Architected a 4-workflow n8n outreach system: daily trigger pulls and filters leads from Google Sheets, injects ICP-specific variables into HTML templates, and sends via Zoho Mail API with OAuth 2.0 refresh, with day-2 and day-3 follow-ups maintaining thread continuity.
- Built an open-tracking webhook writing engagement events and PKT timestamps to the CRM; randomized send delays and batch cooldowns enforce natural sending cadence, eliminating manual outreach effort.

LinkedIn Carousel Automation

n8n · GPT-4o · PDF.co API · Google Drive

- Built a 13-node n8n pipeline that parses PDFs via PDF.co API, generates carousel copy with GPT-4o, and uploads to Google Drive, reducing creation time from hours to under 3 minutes with zero manual steps.

EDUCATION

B.Sc. Computer Science · University of Karachi

Feb 2023 – Present

GPA: 3.0 | Coursework: Software Engineering, AI, Data Structures, Statistics, Computer Networks, Database Systems

Cloud Applied Agentic AI Engineer · Governor Initiative for Artificial Intelligence

Feb 2024 – Present

Hands-on training in TypeScript, Python, and the OpenAI Agents SDK — focused on building production-grade autonomous AI agent systems.

Intermediate in Pre-Engineering · Sir Syed Govt. College for Women

Sept 2020 – June 2022

Percentage: 84.65%

CERTIFICATIONS

IBM RAG and Agentic AI Professional Certificate · IBM / Coursera

- Learned to build end-to-end RAG pipelines, from document ingestion and embedding to vector retrieval and LLM-grounded response generation.
- Built multi-agent systems using LangGraph and CrewAI, learning orchestration patterns like reflection, memory, and conditional logic across coordinated agents.
- Learned to integrate MCP servers to connect, validate, and scale tool-augmented agents in complex AI ecosystems.
- Completed a capstone designing a production-style multimodal RAG system combining structured data, retrieval evaluation strategies, and agentic workflows.

Generative AI: Prompt Engineering Basics · IBM / Coursera

- Learned zero-shot and few-shot prompting techniques to improve LLM output reliability and consistency.
- Practiced advanced prompting methods: Chain-of-Thought, Tree-of-Thought, and Interview Pattern, for producing structured, context-aware responses.
- Explored prompt engineering tools including IBM Watson Prompt Lab, Spellbook, and Dust through hands-on labs.